

# Optimal Planning and Control: Part II

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October 2025

# Continuous-time Optimal Control

Consider the deterministic *continuous-time* dynamical system

$$\frac{d}{dt}x(t) = f(x(t), u(t), t),$$

with initial conditions  $x(0) = x_0$ , state variable  $x(t) \in \mathbb{R}^n$  and control variable  $u(t) \in \mathcal{U} \subseteq \mathbb{R}^m$ .

We will generally assume:

- $f = f(x, u, t)$  to be measurable in  $t$ , continuous in  $u$ , and Lipschitz continuous in  $x$
- $u(t)$  to be piecewise continuous.

# Continuous-time Optimal Control

The **cost function** can be defined as

$$J(x, u) = \int_0^T L(x(t), u(t), t) dt + K(x(T))$$

with running cost  $L(x, u)$  and final cost  $K(x)$ , both  $\mathcal{C}^1$  on their domains.

Solving the **Optimal Control Problem** corresponds to solve the infinite-dimensional optimization

$$J^*(x) = \min_{u \in \mathcal{U}} J(x, u)$$

Plan:

- Hamilton–Jacobi–Bellman approach
- constrained optimization and Pontryagin's Principle

# Continuous-time Optimal Control

The optimal cost-to-go from time  $t$  and state  $x(t)$ ,

$$\begin{aligned} J^*(x, t) &= \min_{u \in \mathcal{U}} \left[ \int_t^T L(x(t'), u(t'), t') dt' + K(x(T)) \right] \\ &= \min_{u \in \mathcal{U}} \left[ \int_t^{t+\Delta t} L(x(t'), u(t'), t') dt' \right. \\ &\quad \left. + \int_{t+\Delta t}^T L(x(t'), u(t'), t') dt' + K(x(T)) \right] \\ &= \min_{\tilde{u}} [L(x(t), \tilde{u}, t) \Delta t \\ &\quad + \min_{\{u\}_{t' \in [t+\Delta t, T]}} \left[ \int_{t+\Delta t}^T L(x(t'), u(t'), t') dt' + K(x(T)) \right]] \\ &= \min_u \{ L(x, u, t) \Delta t + J^*(x, t) \\ &\quad + (\partial_t J^*) \Delta t + (\nabla_x J^*) f(x, u, t) \Delta t \} \end{aligned}$$

# Continuous-time Optimal Control

Reordering terms we get the **Hamilton-Jacobi-Bellman Equation**

$$\partial_t J^*(x, t) + \min_u [L(x, u, t) + (\nabla_x J^*(x, t)) f] = 0$$

with the final condition  $J^*(x(T), T) = K(x(T))$ .

The solution  $\dot{x} = f(x, u^*(x), t)$  is obtained solving the HJB backward in time.

## Remark

HJB is a non-linear PDE: a solution exists, and it is unique, if  $u$  is continuous. If not, solution can exist only in a weak sense (*viscosity solution*).

The optimal cost is defined on  $n + 1$  dimensional space. Discretization is costly (*curse of dimensionality*)

# Constrained optimization formulation

Introduce a modified cost function using Lagrange multipliers

$$\tilde{J} = K(x(T)) + \int_0^T dt \left\{ [L(x(t), u(t), t)] + \lambda(t)^\top (f(x, u, t) - \dot{x}) \right\}$$

and solve the unconstrained optimization for  $\tilde{J}$  by considering the variation w.r.t to all dependent variables  $(x, \lambda, u)$ ,

$$\begin{aligned} \delta \tilde{J} &= (\nabla_x K) \delta x(T) + \int_0^T dt \left\{ \delta L + \lambda^\top (\delta f - \delta \dot{x}) + \delta \lambda^\top (f - \dot{x}) \right\} \\ &= (\nabla_x K) \delta x(T) + \int_0^T dt \left\{ [(\nabla_x L) \delta x + (\nabla_u L) \delta u] \right. \\ &\quad \left. + \lambda^\top [(\nabla_x f) \delta x + (\nabla_u f) \delta u - \delta \dot{x}] + \delta \lambda^\top f - \delta \lambda^\top \dot{x} \right\} \end{aligned}$$

# Constrained optimization formulation

Integrating by parts and reordering terms,

$$\begin{aligned}\delta \tilde{J} = & (\nabla_x K) \delta x(T) - \lambda(T)^\top \delta x(T) + \lambda(0)^\top \delta x(0) \\ & + \int_0^T dt \left[ (\nabla_x L) + \lambda^\top (\nabla_x f) + \dot{\lambda}^\top \right] \delta x(t) \\ & + \int_0^T dt \left[ (\nabla_u L) + \lambda^\top (\nabla_u f) \right] \delta u(t) \\ & + \int_0^T dt \left[ f(x, u, t) - \dot{x} \right]^\top \delta \lambda(t)\end{aligned}$$

where  $\lambda(t)$  is the *adjoint (costate)* variable and  $\delta x(0) = 0$  if  $x(0) = x_0$ .

# Constrained optimization formulation

Setting  $\delta \tilde{J} = 0$ , we get a set of **forward-backward equations**,

From  $t = 0$  to  $t = T$  for  $x(t)$ ,

$$\begin{cases} \dot{x} = f(x, u, t) \\ x(0) = x_0 \end{cases}$$

From  $t = T$  to  $t = 0$  for  $\lambda(t)$ ,

$$\begin{cases} \dot{\lambda} = -(\nabla_x f)^\top \lambda - (\nabla_x L)^\top \\ \lambda(T) = (\nabla_x K|_{t=T})^\top \end{cases}$$

with the additional stationarity condition

$$(\nabla_u L)^\top + (\nabla_u f)^\top \lambda = 0$$

What do they represent?

Is it enough to find the optimal control  $u^*$ , and the optimal trajectory  $x^*$ ?



# Hamiltonian formalism

In **classical mechanics**:

$$\text{Lagrangian } \mathcal{L}(x, \dot{x}) = E_{\text{kin}} - E_{\text{pot}},$$

$$\text{Hamiltonian } \mathcal{H}(x, p) = E_{\text{kin}} + E_{\text{pot}}$$

related by Legendre transform

$$\mathcal{H}(x, p) = p^\top \dot{x} - \mathcal{L}(x, \dot{x})$$

with generalized momenta  $p = (\partial_x \mathcal{L})^\top$ .

## Hamilton's Equations

$$\begin{cases} \dot{x} = (\nabla_p \mathcal{H})^\top \\ \dot{p} = -(\nabla_x \mathcal{H})^\top \end{cases}$$

# Hamiltonian formalism

In control theory: the Lagrangian is replaced by the generalized cost function

$$\mathcal{L}(x, u, \lambda, t) = L(x, u, t) + \lambda^\top (f(x, u, t) - \dot{x}(t))$$

and identifying  $p = -\lambda$  gives the Hamiltonian

$$\begin{aligned}\mathcal{H} &= p^\top \dot{x} - \mathcal{L} \\ &= -\lambda^\top \dot{x} - L(x, u, t) - \lambda^\top f(x, u, t) + \lambda^\top \dot{x}(t) \\ &= -L(x, u, t) - \lambda^\top f(x, u, t)\end{aligned}$$

In the following, we define the Hamiltonian

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^\top f(x, u, t)$$

in order to be consistent with cost minimization.

# Pontryagin's Principle

The corresponding Hamilton's equations describe stationary trajectories

$$\begin{cases} \dot{x} = (\nabla_{\lambda} H)^{\top} = f(x, u, t) \\ \dot{\lambda} = -(\nabla_x H)^{\top} \\ 0 = (\nabla_u H)^{\top} \end{cases}$$

## Theorem (Pontryagin's Minimum Principle)

*If  $u^*(t)$  is the optimal control and  $x^*(t)$  is the corresponding state trajectory, then it exists a costate variable  $\lambda^*(t)$ , such that*

$$\begin{cases} H(x^*, \lambda^*, u^*) = \min_{u \in \mathcal{U}} H(x, \lambda, u) \\ \dot{x}^* = (\nabla_{\lambda} H)^{\top} \Big|_{x^*, \lambda^*, u^*} & x^*(0) = x_0 \\ \dot{\lambda}^* = -(\nabla_x H)^{\top} \Big|_{x^*, \lambda^*, u^*} & \lambda^*(T) = (\nabla_x K)^{\top} \Big|_{x^*(T)} \end{cases}$$

# Pontryagin's (Maximum/Minimum) Principle

If there is a final condition  $\varphi(x(T)) = 0$ , it must be included in the formulation with an additional Lagrange multiplier, i.e.  $+\mu\varphi(x(T))$ . It follows a generalized *transversality condition*

$$\lambda^*(T) = (\nabla_x K)^\top \Big|_{x^*(T)} + \mu^\top (\nabla_x \varphi(x^*(T)))$$

## Remark

PMP gives only *necessary conditions* (first-order conditions) for a given  $x_0$ . Conditions are also sufficient for convex problems. HJB equations always ensure necessary and sufficient conditions (for any initial conditions).

## Remark

HJB equation is a non-linear PDE in  $n + 1$  dimensions (difficult to solve, curse of dimensionality). PMP implies  $2n$  scalar ODE with  $2n$  (non-trivial) mixed boundary conditions (continuity of  $u(t)$  not required).

# Relation between HJB and PMP

Consider  $\dot{x} = f(x, u, t)$  with finite horizon  $T$  and the optimal control law  $u^*(x^*(t), t)$ , then from HJB along the optimal solution, the optimal cost to go must satisfy

$$0 = \partial_t J^*(x^*, t) + L(x^*, u^*, t) + (\nabla_{x^*} J^*) f(x^*, u^*)$$

Differentiating formally w.r.t.  $x^*$ ,

$$\begin{aligned} 0 = \quad & \nabla_{x^*} \partial_t J^* + \nabla_{x^*} L(x^*, u^*, t) + (\nabla_{u^*} L(x^*, u^*, t)) \nabla_{x^*} u^* \\ & + [\nabla_{x^*} f + \nabla_{u^*} f \nabla_{x^*} u^*] \nabla_{x^*} J^* + f^\top (\nabla_{x^* x^*} J^*) \end{aligned}$$

where  $\nabla_{x^* x^*} J^*$  is the Hessian.

On the optimal solution, the quantity  $\nabla_{x^*} J^*$  is in fact only a function of time  $t$ , satisfying

$$\frac{d}{dt} [\nabla_{x^*} J^*] = \nabla_{x^*} (\partial_t J^*) + f(x^*, u^*, t)^\top (\nabla_{x^* x^*} J^*)$$

where we used  $\dot{x}^*(t) = f(x^*, u^*, t)$ .

# Relation between HJB and PMP

Recollecting terms we get

$$\begin{aligned} 0 = & \frac{d}{dt} \nabla_{x^*} J^* + f^\top (\nabla_{x^* x^*} J^*) + \nabla_{x^*} L(x^*, u^*, t) + \nabla_{x^*} J^* \nabla_{x^*} f \\ & + \left\{ \underbrace{(\nabla_{u^*} L(x^*, u^*, t)) + \nabla_{x^*} J^* \nabla_{u^*} f}_{=0} \right\} \nabla_{x^*} u^* \end{aligned}$$

where the term in the brackets vanishes for the HJB minimum condition along the optimal solution. Finally we obtain the equation

$$-\frac{d}{dt} (\nabla_{x^*} J^*(x^*, t)) = \nabla_{x^*} L(x^*, u^*, t) + \nabla_{x^*} J^* \nabla_{x^*} f$$

# Relation between HJB and PMP

Setting  $\lambda(t)^\top = \nabla_{x^*} J^*(x^*, t)$ , we find the conditions that an optimal solution has to satisfy

$$\begin{cases} \dot{x}(t) = f(x, u, t) \\ \dot{\lambda}(t) = - \left[ (\nabla_x L(x, u, t))^\top + \nabla_x f(x, u, t) \lambda(t) \right] \\ u(t) = \arg \min_{\tilde{u}} \left[ L(x, \tilde{u}, t) + f(x, \tilde{u}, t)^\top \lambda(t) \right] \end{cases}$$

with  $\lambda(T) = \nabla_x K(x(T))$  and  $x(0) = x_0$ .

Introducing the Hamiltonian  $H(x, u, \lambda, t) = L(x, u, t) + f(x, u, t)^\top \lambda(t)$ , we get the PMP equations

$$\begin{cases} \dot{x}(t) = (\nabla_u H(x, u, \lambda, t))^\top \\ \dot{\lambda}(t) = -(\nabla_x H(x, u, \lambda, t))^\top \\ u(t) = \arg \min_{\tilde{u}} H(x, \tilde{u}, \lambda, t) \end{cases}$$

# Infinite Time Horizon

**Time Discounting:** in discrete time, the discount factor  $\delta \in (0, 1]$  can be interpreted as the probability that the problem continues to the next time step.

Introduce a stopping rate  $\rho$ , then the problem continues with probability  $1 - \rho\Delta t$ .

After  $n = t/\Delta t$  steps,

$$(1 - \rho\Delta t)^n = (1 - \rho\Delta t)^{t/\Delta t} \xrightarrow{\Delta t \rightarrow 0} e^{-\rho t}$$

Define the discounted cost function

$$J(x, u) = \int_0^T e^{-\rho t} L(x, u, t) dt + e^{-\rho T} K(x(T)).$$

and the cost-to-go

$$J(x, u, t) = \int_t^T e^{-\rho t'} L(x, u, t') dt' + e^{-\rho T} K(x(T)).$$



# Infinite Time Horizon

The optimal cost-to-go  $J^*(x, t) = \min_u J(x, u, t)$  solves the **(time-discounted) HJB equation**

$$\rho J^*(x, t) = \min_{u \in \mathcal{U}} \{L(x, u, t) + \partial_t J^*(x, t) + (\nabla_x J^*(x, t)) f(x, u, t)\}$$

with terminal condition  $J^*(x, T) = e^{-\rho T} K(x(T))$ .

Limit  $T \rightarrow \infty$ : assuming time invariance (i.e.  $L = L(x, u)$ ,  $f = f(x, t)$ ), the HJB equation reduces to

$$\rho J^*(x) = \min_{u \in \mathcal{U}} \{L(x, u) + (\nabla_x J^*(x)) f(x, u)\}$$

and, in the absence of discounting, it further simplifies into

$$\min_{u \in \mathcal{U}} \{L(x, u) + (\nabla_x J^*(x)) f(x, u)\} = 0.$$

# Linear Quadratic Control

Consider the linear matrix equation

$$\dot{x}(t) = Ax(t) + Bu(t)$$

with initial condition  $x(0) = x_0$ .

Consider the quadratic cost

$$J = \int_0^T \frac{1}{2} \left( x(t)^\top Q x(t) + u(t)^\top R u(t) \right) dt + \frac{1}{2} x(T)^\top M x(T)$$

where  $Q, M \in \mathbb{R}^{n \times n}$  (symm. non-negative semidefinite) and  $R \in \mathbb{R}^{m \times m}$  (symm. positive definite).

Define the Hamiltonian

$$H(x, u, \lambda) = \frac{1}{2} (x^\top Qx + u^\top Ru) + \lambda^\top (Ax + Bu)$$

then the optimality conditions are

$$\begin{cases} \dot{x} = (\nabla_\lambda H)^\top = Ax + Bu \\ \dot{\lambda} = -(\nabla_x H)^\top = -Qx - A^\top \lambda \\ u = -R^{-1}B^\top \lambda \end{cases}$$

with  $x(0) = x_0$  and  $\lambda(T) = Mx(T)$ .

How can we find a solution?

# Linear Quadratic Control

It is convenient to introduce a time-varying matrix  $S(t)$ , s.t.

$$\lambda^*(t) = S(t) x^*(t)$$

so that the optimal condition can be written in *linear feedback* form

$$u^*(t) = -R^{-1}B^\top S(t) x^*(t).$$

Then

$$\dot{\lambda}(t) = -Qx(t) - A^\top S(t)x(t)$$

but also

$$\begin{aligned}\dot{\lambda}(t) &= \dot{S}(t)x(t) + S(t)\dot{x}(t) \\ &= \dot{S}(t)x(t) + S(t)[Ax(t) + Bu(t)]\end{aligned}$$

# Linear Quadratic Control

Under optimality, the matrix  $S(t)$  solves the **continuous-time Riccati equation**

$$\dot{S} = -Q - A^T S - SA + SBR^{-1}B^T S$$

with  $S(T) = M$ .

## Proposition

*The Riccati ODE has a unique solution and,  $\forall t \in [0, T]$ , the solution is symmetric non-negative semidefinite.*

## Proof.

The existence and uniqueness follow from local Lipschitz property of the r.h.s. of the Riccati ODE. Then show that  $d/dt (S - S^T) = 0$  only on the solution for which  $S^T = S$ . □

# Linear Quadratic Control

## Definition

A (LTI) system  $\dot{x} = Ax + Bu$  is **controllable** if for every initial state  $x_0$  and final state  $x_T$ , there exists a time  $T$  and control  $u(t)$  such that  $x(0) = x_0$  and  $x(T) = x_T$ .

## Proposition

The LTI system  $\dot{x} = Ax + Bu$  is controllable if the **controllability** ( $n \times nm$ ) **matrix**  $\mathcal{C}$  defined as

$$\mathcal{C} = [B, AB, A^2B, \dots, A^{n-1}B]$$

is full rank (rank  $n$ ).

Controllability can be defined for more general time-dependent  $A, B$  and finite time horizon.

# Linear Quadratic Control

*Sketch of the proof:* A formal solution of the LTI system is

$$x(t) = e^{At}x(0) + \int_0^t dt' e^{A(t-t')} Bu(t')$$

Setting  $x_0 = 0$  for simplicity and using Cayley–Hamilton theorem (i.e.  $e^{At} = \sum_{\ell=0}^{n-1} \alpha_\ell(t) A^\ell$ , where  $\alpha_\ell(t)$  are computable scalars),

$$x(t) = \sum_{\ell=0}^{n-1} A^\ell B \int_0^t dt' \alpha_\ell(t-t') u(t') = \sum_{\ell=0}^{n-1} A^\ell B w_\ell(t)$$

where  $w$  are  $m$ -dimensional vectors. For instance,

$$\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = [B, AB, A^2B] \begin{bmatrix} w_0(t) \\ w_1(t) \\ w_2(t) \end{bmatrix}$$

The state  $x(t)$  can be controlled if changing the two inputs (and then  $w_\ell(t)$ ) we can steer  $x(t)$  where desired. This is possible if the matrix  $[B, AB, A^2B]$  is full rank.

## Theorem

For a controllable linear time-invariant system  $\dot{x} = Ax + Bu$  with cost

$$J = \frac{1}{2} \int_0^\infty [x^\top Qx + u^\top Ru] dt,$$

with  $Q, M \in \mathbb{R}^{n \times n}$  (symm. non-negative semidefinite) and  $R \in \mathbb{R}^{m \times m}$  (symm. positive definite), the optimal control is

$$u^*(x) = -R^{-1}B^\top Sx$$

where  $S$  is the symmetric non-negative semidefinite solution of the **Algebraic Riccati equation**

$$SA + A^\top S + Q - SBR^{-1}B^\top S = 0.$$



# Linear Quadratic Control

## Definition

Consider an unobserved LTI system  $\dot{x} = Ax$ , and observations  $y = Cx$ , where  $y \in \mathbb{R}^m$ ,  $C \in \mathbb{R}^{m \times n}$ . The **observability** ( $mn \times n$ ) **matrix**  $\mathcal{O}$ :

$$\begin{cases} y = Cx \\ \dot{y} = C\dot{x} \\ \vdots \\ y^{(n-1)} = CA^{n-1}x \end{cases} \implies Y = \mathcal{O}x \quad \mathcal{O} = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

The system is *observable* if matrix  $\mathcal{O}$  is full rank (again  $n$ ).

Controllability and Observability are dual notions:

- one can observe the past but not control it
- one can control the future but not observe it

# Stochastic control in a nutshell

Consider the discrete-time stochastic process in which

$$x_{k+1} \sim \mathcal{N}(x_k + f_k(x_k, u_k) \Delta, \Sigma(x_k, u_k) \Delta)$$

at time  $t = k\Delta$  and with covariance matrix

$$\Sigma(x, u) = \sigma(x, u) \sigma(x, u)^\top.$$

## Remark

Why mean and variance are  $O(\Delta)$ ? Consider a random walk with unit step, drift  $f$  and std  $\sigma$ , after  $k$  steps the root mean square displacement is  $\bar{R}_k \sim (f^2 k^2 + \sigma^2 k)^{1/2}$ . In the continuous-time limit we have Brownian motion (*Wiener process*). This is the elementary brick to build continuous-time stochastic control.

# Stochastic control in a nutshell

Starting from the Bellman equation for discrete-time optimal planning

$$V_k(x_k) = \max_{a_k} \{r_k(x_k, a_k) + \mathbb{E}_{w_{k+1}} [V(f_k(x_k, a_k, w_{k+1}))]\}$$

and replacing action  $a_k$  by control  $u_k$ , reward  $r_k(x, a)$  by running cost  $L_k(x, u) \Delta$ , specifying the new disturbancy  $w_{k+1}$ , replacing the optimal value  $V_k(x)$  by optimal cost  $J_k(x)$  and maximization by minimization, we get

$$J_k(x_k) = \min_{u_k} \{L_k(x_k, u_k) \Delta + \mathbb{E}_{w_{k+1}} [J_{k+1}(x_k + f_k(x_k, u_k) \Delta + w_{k+1})]\}$$

with  $w_{k+1} \sim \mathcal{N}(0, \Sigma(x_k, u_k) \Delta)$  and terminal cost  $J_T(x) = K(x)$ .

# Stochastic control in a nutshell

Performing a second-order expansion of  $J$ , we get

$$J(x + \delta_x) = J(x) + (\nabla_x J(x)) \delta_x + \frac{1}{2} \delta_x^\top \nabla_{xx} J(x) \delta_x + O(\delta_x^3)$$

with  $\delta_x = f_k(x_k, u_k) \Delta + w_{k+1}$ .

Collecting terms up to first order in  $\Delta$ ,

$$\begin{aligned} & \mathbb{E}_{w_{k+1}} [J_{k+1}(x_k + f_k(x_k, u_k) \Delta + w_{k+1})] \\ & \approx J_{k+1}(x_k) + (\nabla_x J_{k+1}(x)) f_k(x_k, u_k) \Delta + \frac{1}{2} \mathbb{E} \left[ w_{k+1}^\top \nabla_{xx} J_{k+1}(x_k) w_{k+1} \right] \\ & = J_{k+1}(x_k) + (\nabla_x J_{k+1}(x)) f_k(x_k, u_k) \Delta + \frac{1}{2} \text{tr} [\nabla_{xx} J_{k+1}(x_k) \Sigma(x_k, u_k)] \Delta \end{aligned}$$

$$\text{using } \mathbb{E} [w^\top A w] = \mathbb{E} [\text{tr}(w^\top A w)] = \text{tr}(A \mathbb{E}[w w^\top]) = \text{tr}(A \Sigma)$$

# Stochastic control in a nutshell

The discrete-time Bellman equation at  $O(\Delta)$  becomes

$$\frac{J_k(x_k) - J_{k+1}(x_k)}{\Delta} = \min_{u_k} \left\{ L_k(x_k, u_k) + (\nabla_x J_{k+1}(x)) f_k(x_k, u_k) + \frac{1}{2} \text{tr} [\nabla_{xx} J_{k+1}(x_k) \Sigma(x_k, u_k)] \right\}$$

Taking the continuous-time limit ( $\Delta \rightarrow 0$ )

$$-\partial_t J(x, t) = \min_{u \in \mathcal{U}} \left\{ L(x, u, t) + (\nabla_x J) f(x, u, t) + \frac{1}{2} \text{tr} (\Sigma(x, u) \nabla_{xx} J) \right\}$$

which is the Hamilton-Jacobi-Bellman equation for the stochastic control problem described by the continuous-time (Ito) *stochastic differential equation*

$$dx = f(x, u, t) dt + \sigma(x, u) dw$$

where  $dw(t)$  is a Wiener process (with zero mean and variance  $dt$ ).